DHRUV CHHABRA

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Work Experience

Quantitative Trader - Commodities; Alpha Alternatives Pvt Ltd — May. 2024 – Sept. 2025

- **Multi-Asset Pairs Trading & Risk Framework** – Using `Python`
  - Collaborated with discretionary traders to formulate a **early mean-reversion signal** strategy
  - Used gradient-based momentum to capture trend reversal, with **volatility-adjusted stop-loss** to prevent tail-risk
  - Achieved an overall Profit Factor of 1.9 and 45% win rate on portfolio of ₹6 Cr, **generating ₹20 lakh+** in year 2025 alone
- **Algorithmic Strategy Refinement and Portfolio Optimisation system** – Using `Python`
  - Led a **strategic audit** of a trend-following algorithm for commodities' Futures, rectifying critical backtesting biases
  - Automated signal generation framework, **reducing operational errors and slippage cost** from 0.15% to 0.02%
  - Applied Markowitz Mean-Variance theory to rebalance portfolio weights, **boosting monthly returns** to 1.38%
- **Quantitative Strategy Research and Prototyping** – Using `Python`
  - Finding an alpha on Panama backwards adjusted Spreads data by building on the mean reversion principle
  - Built an XGBoost model using economic features and seasonality analysis to predict the ideal allocation size

Associate Data Scientist; Optum Global Solutions Ltd — Aug. 2021 – May 2024

- **Automated Clinical Documentation & Validation System** – Using `Python`
  - Developed a Gen AI engine to automate medical claim appeal summaries, achieving **73% coverage** of all appeal categories
  - Built a correctness evaluation ML model to score AI outputs, **reducing manual audit requirement** by 85%
  - Designed a **double-blind validation tool** to collect data and benchmark AI fidelity against clinician-written summaries
  - Validated **AI non-inferiority** via 2-sample T-test, **reducing documentation time** by 55–60%
- **Prediction of Risk of Hospital Readmission** – Using `Python & Pyspark`
  - Worked on a 30-day readmission problem to identify high-risk patients and reduce avoidable hospital costs
  - Built a risk-scoring model on an imbalanced population (**23% incidence**), achieving actionable predictive performance
  - Presented **cost to benefit analysis** for a specific use case to quantify model's utility to senior leadership

Projects

The Mental Health Coverage Gap in Indian Health Insurance — Self | May 2026

- Quantified gap in India's mental health insurance: even best-in-class retail policies cover close to 0% of outpatient therapy
- Proposed B2B benefits platform redesign with hybrid funding to close the gap that retail insurance economically cannot

Strategic Roadmap: Protecting Margins Amidst Global Silver Shortages — Self | Apr. 2026

- Identified structural EBITDA risk for Solar PV manufacturers from silver supply constraints and stress-tested margin impact
- Developed a strategic response with financial hedging, direct offtake agreements with primary miners, and silver intensity R&D, to shift manufacturers from variable to predictable cost structures

Algorithmic Trading with Backtesting Analysis – Using `Python` — Self | Jan. 2024

- Developed systematic trading strategies (alphas) using technical analysis indicators to generate buy/sell signals
- Back-tested the results against S&P500 as the benchmark and computed the draw-downs and risk-adjusted returns

Option Pricing for an FX Derivative – Using `C++` — Self | Dec. 2023

- Implemented the Black-Scholes Model to price an FX call option using: the analytical solution, finite difference method, and Monte Carlo simulation for asset prices with discretized time intervals
- Analysed the option Greeks like Delta, Gamma and Vega, and plotted the parabolic option prices for each method

Time Series Analysis of stock price movement – Using `R` — Prof. A Mitra | Dec. 2020

- Conducted tests for stationarity and seasonality, and evaluated ACF and PACF plots to select order for the model
- Fitted an ARIMA model on transformed data and conducted residual analysis to verify random distribution of errors

Skills

`Advanced Excel (VBA)`, LaTeX*, `Git, SPSS, OpenAI, Python*, SQL*, C/C++`

Education

MSc, Statistics	Indian Institute of Technology Kanpur	8.33 /10	July 2019 - May 2021
BSc, Statistics (Hons.)	PGDAV College, University of Delhi	9.47 /10	July 2016 - May 2019

Key Courses

Financial Statistics | Statistical Methods for Business Analytics | Statistical Simulation & Inference | Econometrics

Extra Curricular

- Awarded the **Junior Diploma** in Hindustani Classical Music **(Violin)** from Prayag Sangeet Samiti, 2025
- Qualified the **Goethe-Zertificat A2** level **German** language examination from Goethe-Institut Delhi, 2024
- Awarded **1st Best Speaker** in **Novice Debating League** organised by DebSoc IIT Kanpur, 2020